

HOMEWORK 5 FOR MATH 2860 (ELEMENTARY DIFFERENTIAL EQUATIONS)

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Due at 10:00 EST in class

Question Q1. A tank initially contains 2,000 litres of water with a concentration of mercury of 6.0×10^{-5} grams per litre. Water that contains 3.0×10^{-5} grams per litre is pumped into the tank at 100 litres per hour. The well mixed solution is pumped out of the tank at 100 litres per hour. How long will it take for the concentration in the tank to reach 4.5×10^{-5} grams per litre?

Question Q2. A vat with capacity of 500 gallons initially has 10 lbs. of salt dissolved in 100 gallons of brine. A brine solution with 0.5 lbs. of salt per gallon is pumped into the tank at a rate of 4 gallons per minute, and the well mixed solution is pumped out at a rate of 2 gallons per minute. How much salt will be in the vat when the vat is full? (Hint: first find the solution for the volume of the fluid in the tank, before tackling the amount of salt in it.)

Question Q3. A car's fuel tank holds 15 gallons of gasoline. When the tank is full, the gasoline contains 0.015 gallons of ethanol. Suppose that pure gasoline is pumped into the tank at a rate of 1 gallon per minute, and the well mixed solution is pumped out at a rate of 1 gallon per minute. How long will it take for the amount of ethanol in the tank to be 0.003 gallons?

Show all your work.